# Experiment No.: 7

**1. Aim:-** Design the architecture and implement the autoencoder model for Image Compression.

**2. Objective:-** to design and implement an autoencoder model for image compression using the MNIST dataset**.** The model will compress images into a lower-dimensional representation and reconstruct them, evaluating its effectiveness by comparing the original and reconstructed images**.**

**3. Outcome:-**

**4. Theory**

**Procedure**

Step 1: Import Libraries

Step 2: Load and Preprocess the Data

- Load the MNIST dataset.
- Normalize the image data between 0 and 1**.**
- Flatten the images into 1D arrays for the autoencoder model.

### Step 3: Build the Autoencoder Model

- The encoder reduces the image dimensions (compression).
- The decoder reconstructs the image from the lower-dimensional encoding.

### Step 4: Train the Autoencoder

- Train the model with **20 epochs** and a batch size of 256.
- Use the training data for both input and target to learn compression and reconstruction.

### Step 5: Evaluate the Model

- Use the model to compress and reconstruct test images.
- Plot the original and reconstructed images to evaluate the quality of compression.

**5. Results:**

**Compression Efficiency:**

- The autoencoder effectively compresses the images into a lower-dimensional representation of 64 dimensions from the original **784 dimensions**.

**Reconstruction Quality:**

- The reconstructed images retain the main features of the original images, although some details may be slightly blurred due to compression loss.

**Loss Reduction:**

- The **binary cross-entropy loss** decreases consistently across the epochs, indicating that the model learns to compress and reconstruct efficiently.

**Visualization:**

- The reconstructed images closely resemble the original images, demonstrating the model's effectiveness in **image compression**.

## Conclusion

The autoencoder model successfully performed image compression on the MNIST dataset**,** reducing the image dimensions while preserving essential features. The reconstructed images closely resembled the original ones, demonstrating the model's ability to retain visual quality with minimal information loss. These practical highlights the effectiveness of autoencoders for dimensionality reduction and image compression tasks